Handout #18
Thursday, 15 October 2026




Autumn Quarter 2026

# Problem Set 4

**Due:** Tuesday, 27 October 2026 at 1:30pm via Gradescope
**Topics:** Satellite communications, interplanetary mission analysis

Computer Files Provided for this Problem Set

You can find the following MATLAB and supporting files on the AA 279B Canvas website in the "Problem Sets and Solutions" folder:

| | |
|---|---|
| **`AA279lambert_curtis.m`** | Curtis's Lambert solver using Universal Variables, slightly modified. Note in the comments that this accepts **`'pro'`** (prograde) or **`'retro'`** (retrograde) as an input. |
| **`AA279j2000_planetary_elements.m`** | Returns planetary orbital elements and rates for requested planet. |
| Galileo Mission Overview - NASA.pdf | Overview of the Galileo mission, with flyby altitude data. |
| Galileo Trajectory - NASA.png | Plot of Galileo's interplanetary trajectory. |

**4.1)** (25 points total) This assignment challenges you to reverse-engineer the trajectory flown by NASA's Galileo mission launched to Jupiter in 1989. For this problem, you will use ephemeris data to determine the location of Earth, Venus, and Jupiter at key times during the mission. You will then plot these locations along with that of Galileo's encounter with the asteroid Gaspra.

a) (12 points) Create a function that returns the heliocentric position (in [km]) and velocity (in [km/sec]) of a planet given the planet ID number (1 for Mercury through 9 for Pluto) and the Julian date. You have been provided with a MATLAB function called **`AA279j2000_planetary_elements.m`** that returns raw planetary orbital element and element rate data. Note that this function returns semi-major axis and its derivative in [AU] and [AU/century]; convert these using 1 [AU] = 149,597,870.7 [km].
Also, $\mu_{sun}$ = 1.3271244004193938x$10^{11}$ [$km^3/sec^2$].

b) (13 points) The following table lists (with some simplifications) the dates and times when NASA's Galileo spacecraft left Earth orbit and made subsequent swingbys of various celestial bodies on its way to arrival at Jupiter:

| **Waypoint** | **Date** | **Time [hours:minutes:seconds]** |
|---|---|---|
| Earth Departure | 26 October 1989 | 00:55:00 UTC |
| Venus Swingby | 10 February 1990 | 05:59:00 UTC |
| Earth Swingby #1 | 8 December 1990 | 20:35:00 UTC |
| Gaspra Encounter | 29 October 1991 | 22:49:00 UTC |
| Earth 2 Swingby #2 | 8 December 1992 | 15:09:00 UTC |
| Jupiter Arrival | 7 December 1995 | 21:53:44 UTC |

source: NASA Goddard's NSSDC website at
http://nssdc.gsfc.nasa.gov/nmc/spacecraftOrbit.do?id=1989-084B

Although the **`AA279j2000_planetary_elements.m`** function doesn't supply information for Gaspra, NASA JPL's HORIZONS website lists Gaspra's heliocentric coordinates at its encounter time as:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} +7.635070746360724\text{E}+07 \\ -3.201967071615438\text{E}+08 \\ +1.175159636590249\text{E}+07 \end{bmatrix} [km]$$

Make a single plot (on heliocentric *x* and *y* axes labeled in [AU]) of the positions of Earth, Venus, Jupiter, and Gaspra corresponding to the times listed in the table above. To facilitate later comparison with NASA's official data, your plot should have the positive x axis pointing to the *right* on the page and the positive y axis pointing *up*. Earth should be plotted at only three positions (corresponding to the three "Earth" times in the table), Venus at one position, Gaspra at one position, and Jupiter at one position. Also show the position of the Sun at the origin.

**4.2)** (45 points total) Based on only the dates and positions above, for this problem you will estimate some of the important parameters of the Galileo mission and plot its trajectory.

Note About Lambert Solvers

Since interplanetary trajectories sometimes reach solar escape speeds, we can't ignore parabolic/hyperbolic trajectories as we did when analyzing a system confined to LEO trajectories. Also, some of Galileo's trajectory segments span less than 180° of true anomaly (short-way) and some span more (long-way). Here are two suggestions for getting robust Lambert solver results for interplanetary missions:

- Use the **`AA279lambert_curtis.m`** function provided to you. This is adapted from a function provided with *Orbital Mechanics for Engineering Students* by Howard Curtis. Look closely at the first 10 lines of comments documenting the inputs. Whereas Vallado's solvers accept **`'s'`** and **`'l'`** for short-way and long-way, Curtis's solver instead accepts **`'pro'`** and **`'retro'`** for prograde (positive around the *z* axis according to the right hand rule) and retrograde (opposite of prograde). This is an important distinction: essentially all the Lambert solutions we will use for interplanetary trajectories will be prograde, even if the transfer angle exceeds 180°. As long as we know the transfer is prograde, using Curtis's solver with **`'pro'`** avoids having to decide between **`'s'`** and **`'l'`**.

- Use **`AA279lambert_vallado_u.m`** (Vallado's universal variables solver already provided) with some additional logic that decides between short-way and long-way solutions before the function call.

Both solvers seem to produce good results for this problem; we encourage you to try both.

Planetary Data

| Planet | Gravitational Parameter μ [km$^3$/sec$^2$] | Mean Volumetric Radius [km] |
|---|---|---|
| Venus | 324,858.630 | 6,051.80 |
| Earth | 398,600.440 | 6,371.01 |
| Jupiter | 126,686,511.000 | 69,911.00 |

source: JPL's HORIZONS website at http://ssd.jpl.nasa.gov/horizons.cgi

a) (24 points) For each of the three planetary swingbys (Venus, Earth #1, and Earth #2), write code to find the following:

- heliocentric asymptotic arrival and departure velocity vectors, using Lambert
- planetocentric asymptotic arrival and departure velocity vectors (although these velocity vectors are measured relative to the swingby planet's reference frame, they can be left expressed in heliocentric coordinates)
- planetocentric asymptotic arrival and departure speeds $\left|{}^{planet}\vec{v}_{-\infty}\right| \; and \; \left|{}^{planet}\vec{v}_{+\infty}\right|$
- hyperbolic passage turning angle $\delta_{planet}$
- hyperbolic passage eccentricity $e_{planet}$

You don't need to report all of these quantities; just tell us what relationships you used to write the code. Although one segment of the trajectory ends at Gaspra and another begins there, you don't need to compute the values above for the Gaspra encounter since it barely affected the magnitude or direction of Galileo's velocity vector.

b) (3 points) For each of the three planetary swingbys, what are the planetocentric asymptotic arrival and departure speeds, and are they nearly equal (say to within 4%)? (These two speeds should be equal for an *ideal* unpowered swingby with no added ΔV. Don't worry if they don't match exactly.)

c) (8 points) For each of the three planetary swingbys, what is the hyperbolic passage eccentricity and approximate closest approach distance (periapsis radius minus mean volumetric radius)? If you need to compute $v_\infty^2$, a good approximation is:

$$^{planet}v_\infty^2 = \left|{}^{planet}\vec{v}_{-\infty}\right|\left|{}^{planet}\vec{v}_{+\infty}\right|$$

d) (10 points) Use MATLAB's **`ode113`** propagator with **`options = odeset('RelTol', 1e-6, 'AbsTol', 1e-9)`** to simulate the trajectory in five segments (Earth-to-Venus, Venus-to-Earth#1, Earth#1-to-Gaspra, Gaspra-to-Earth#2, and Earth#2-to-Jupiter). On each segment, assume only the Sun's gravity is active. Plot the segments on heliocentric x and y axes labeled in [AU], with the x and y axes oriented as before. On the same plot, show the Sun at the origin and all the planet and asteroid locations on their respective departure, swingby, encounter, or arrival dates. At the scale of this plot, does the end of each segment look like it matches up with the beginning of the next?

**4.3)** (18 points total) Now you'll determine how some simple communications parameters evolved during the mission.

a) (6 points) Write code that steps through each point in your assembled trajectory and finds distance from Earth's center (which moves and can be determined from ephemeris data) to Galileo. At each point, also compute spreading loss in [dB 1/m$^2$]. This is $L_{spread} = \frac{1}{4\pi R^2}$ (undefined when $R = 0$), or more usefully for our purpose:

$$L_{spread} = -70.99 - 20\log_{10} R_{km} \qquad \text{in } \left[\text{dB}\frac{1}{\text{m}^2}\right]$$

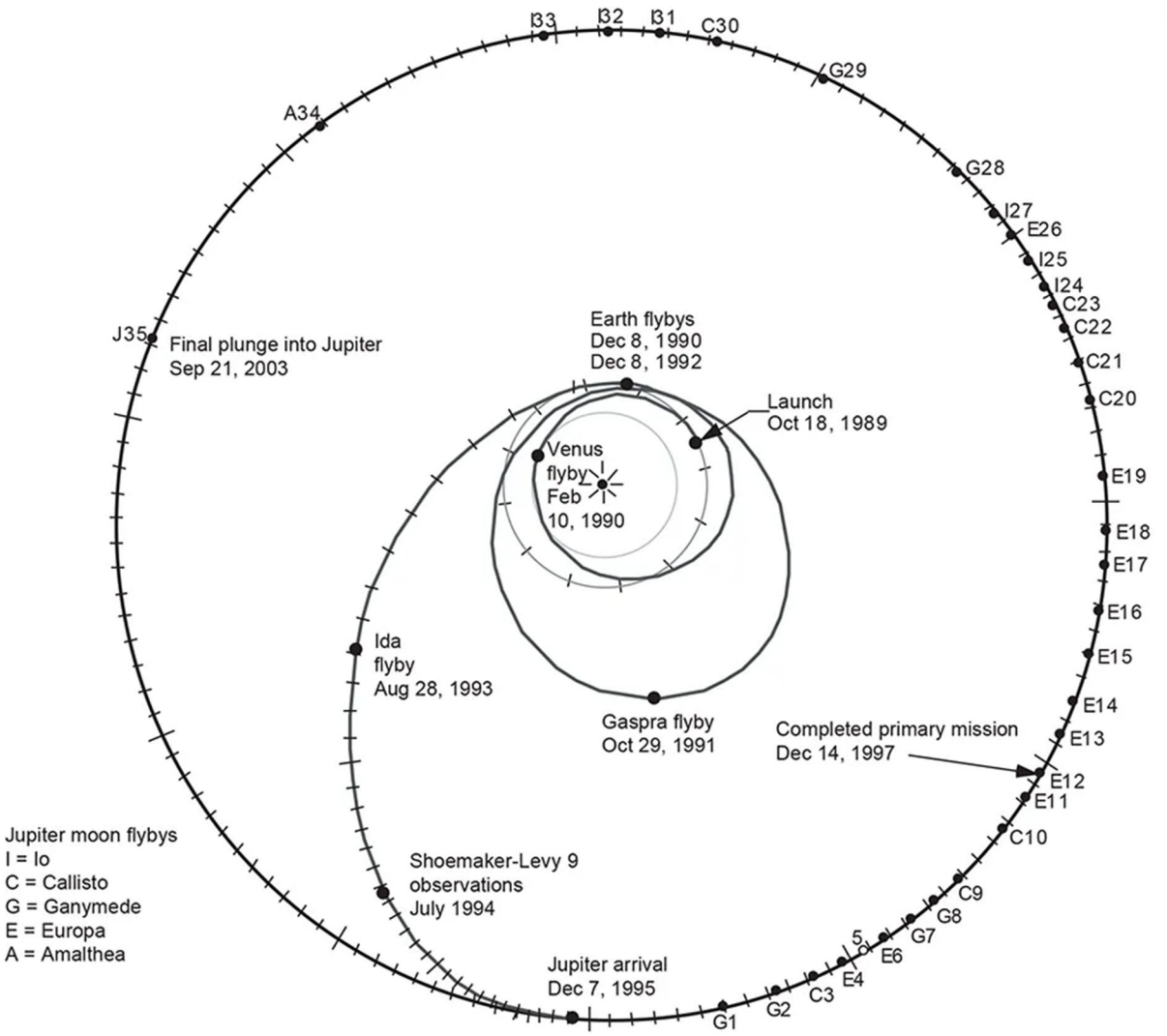
where $R_{km}$ is the Earth-probe distance in [km].

b) (6 points) Plot Earth-probe distance in [AU] versus time elapsed since Earth departure in [years].

c) (6 points) Plot spreading loss in [dB 1/m$^2$] versus time elapsed since Earth departure in [years].

**4.4)** (12 points total) Finally, discuss how your results compare with publicly-available data from the following NASA sources and comment on any differences you see:

| **Your Results** | **Comparison Source** |
|---|---|
| swingby eccentricities | Venus Swingby hyperbolic eccentricity: 3.59<br>Earth Swingby #1 hyperbolic eccentricity: 2.47<br>Earth Swingby #2 hyperbolic eccentricity: 2.32<br><br>source:<br>http://nssdc.gsfc.nasa.gov/nmc/spacecraftOrbit.do?id=1989-084B<br>(NOTE: some of the swingby periapsis data given there are incorrect!) |
| swingby closest approach distances | Venus Swingby altitude: 16,000 [km]<br>Earth Swingby #1 altitude: 960 [km]<br>Earth Swingby #2 altitude: 303 [km]<br><br>source: Galileo Mission Overview - NASA.pdf |
| trajectory plot | trajectory plot below from Galileo Trajectory - NASA.png |




*Trajectory and key mission events*

Galileo Trajectory